

A/B Test for New AI Model Deployment in Loan Approval Decisions

A/B testing

Risk analytics

Executive recommendation

R

A business analytics case on whether the new AI decision-support model should be applied in the loan approval workflow.

The company needs evidence on whether the new AI model improves real lending decisions.

Why This Matters ?

Existing Decision Support System

Legacy Model

- Deployed two years ago to support loan officers.
- Economic conditions have changed, potentially reducing model effectiveness.

Business Challenge

Costly loan approval mistakes remained

Type I Error → Rejecting good loans (lost revenue opportunities)
Type II Error → Approving bad loans (credit losses and defaults)

Business Response

New AI Model Developed

Expected to improve recommendation quality and reduce costly lending errors.

Loan approval workflow

1 Initial Human Review

Officer approves or rejects applications without seeing model prediction.

2 Model Recommendation

System shows either legacy or new model prediction, depending on variant.

3 Final Decision

Officer may revise the decision after considering model advice.

4 Outcome Evaluation

Final errors are assessed against actual loan outcome.

Experiment: AB Test

Control

Legacy Model

Treatment

new AI model

Analytical Challenge

Randomized groups were established, but the experiment **lacked predefined hypotheses and Overall Evaluation Criteria (OECs)**.

The deployment decision was decomposed into three analytical dimensions

Business outcomes, model performance, and human-AI collaboration — each dimension asks a different question.

Business Value of the New AI Model					
Good Loan Perspective		Bad Loan Perspective		Human-Model Collaboration Mechanism	
A1	Model Accuracy AI Type I Error: Legacy Model vs. New Model	B1	Model Accuracy AI Type II Error: Legacy Model vs. New Model	C1	Adoption Rate Agreement/Conflict Metrics with model
A2	Initial Human Decision Type I Initial: Baseline comparison between groups	B2	Initial Human Decision Type II Initial: Baseline comparison between groups	C2	Confidence and Productivity confidence_fin / complt_fin
A3	Final Human + Model Decision Type I Final: Core experimental outcome	B3	Final Human + Model Decision Type II Final: Core experimental outcome		
A4	Improvement Magnitude Improvement from A2 → A3	B4	Improvement Magnitude Improvement from B2 → B3		

Because default losses are the primary business risk, Type II errors become the key success metric

The primary OEC measures business impact, while supporting OECs explain model quality, decision improvement, and model adoption.

Evidence Area	Key Evaluation Question	OEC	Direction of Success
Business Outcome	Does deploying the new AI model lead to better final lending decisions?	Primary Final Type II Rate Final Type II errors / Total bad loans	↓
Model Performance	Does the new AI model provide better recommendations?	Supporting Model Type II Error Rate Model Type II errors / Total bad loans	↓
Decision Improvement	Does AI model assistance help officers reduce Type II errors?	Supporting Type II Reduction Initial Type II Rate – Final Type II Rate	↑
Human - Model Collaboration	Are officers more likely to adopt AI recommendations in their final decisions?	Supporting Agreement Lift Final Agreement Rate – Initial Agreement Rate	↑
		Supporting Conflict Reduction Initial Conflict Rate – Final Conflict Rate	

From daily workflow logs to officer-level evidence

Daily workflow records were cleaned and aggregated into officer-level metrics to align the unit of analysis with the randomized A/B test design.

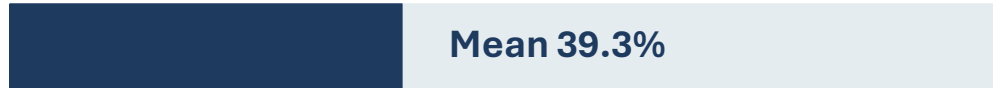
Original Data Structure	Officer × Day Each row represents one officer observed on one day. <table><tr><th>officer_id</th><th>Day</th><th>Variant</th><th>Model_Typell</th><th>Typell_init</th><th>Typell_fin</th><th>agree_init</th><th>agree_fin</th><th>conflict_init</th><th>conflict_fin</th><th>fully_complt</th><th>badloans_num</th><th>goodloans_num</th></tr><tr><td>qamcqdoe</td><td>1</td><td>Treatment</td><td>2</td><td>2</td><td>2</td><td>7</td><td>10</td><td>2</td><td>0</td><td>9</td><td>4</td><td>6</td></tr><tr><td>qamcqdoe</td><td>2</td><td>Treatment</td><td>2</td><td>3</td><td>3</td><td>8</td><td>8</td><td>2</td><td>2</td><td>10</td><td>5</td><td>5</td></tr><tr><td>2udooyt</td><td>1</td><td>Control</td><td>2</td><td>0</td><td>0</td><td>3</td><td>7</td><td>1</td><td>1</td><td>4</td><td>4</td><td>6</td></tr><tr><td>2udooyt</td><td>2</td><td>Control</td><td>2</td><td>3</td><td>2</td><td>8</td><td>9</td><td>2</td><td>1</td><td>10</td><td>5</td><td>5</td></tr></table>												officer_id	Day	Variant	Model_Typell	Typell_init	Typell_fin	agree_init	agree_fin	conflict_init	conflict_fin	fully_complt	badloans_num	goodloans_num	qamcqdoe	1	Treatment	2	2	2	7	10	2	0	9	4	6	qamcqdoe	2	Treatment	2	3	3	8	8	2	2	10	5	5	2udooyt	1	Control	2	0	0	3	7	1	1	4	4	6	2udooyt	2	Control	2	3	2	8	9	2	1	10	5	5	190 Control observations In original data	280 Treatment observations In original data
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Data Cleaning	Incomplete human-model review workflows were excluded because the final decision outcome could not be observed. ▶ Removed records with fully_complt = 0	100 Control observations after cleaning	280 Treatment observations after cleaning																																																																												
	Data Aggregation			Creates one independent observation per officer for valid statistical comparison. ▶ Summed Type II errors, agreement, conflict, revision, and confidence metrics across each officer's 10-day period.	10 Loan Officers in Control	28 Loan Officers in Treatment																																																																									
				Makes performance comparable across officers regardless of review volume. ▶ Converted counts into rates using actual bad-loan and good-loan totals.																																																																											

The introduction of the new AI model significantly reduced the final Type II error rate.

Treatment officers had a lower Final Type II Rate than Control officers, with both statistical and practical significance.

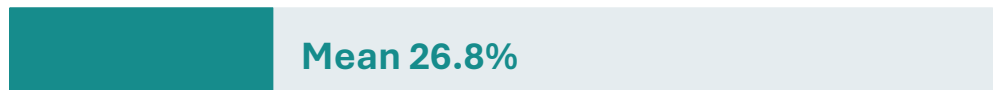
Primary OEC: Final Type II Rate

Control: legacy model



Mean 39.3%

Treatment: new AI model



Mean 26.8%

**- 12.5
pp**

final Type II rate gap

Welch's t-test

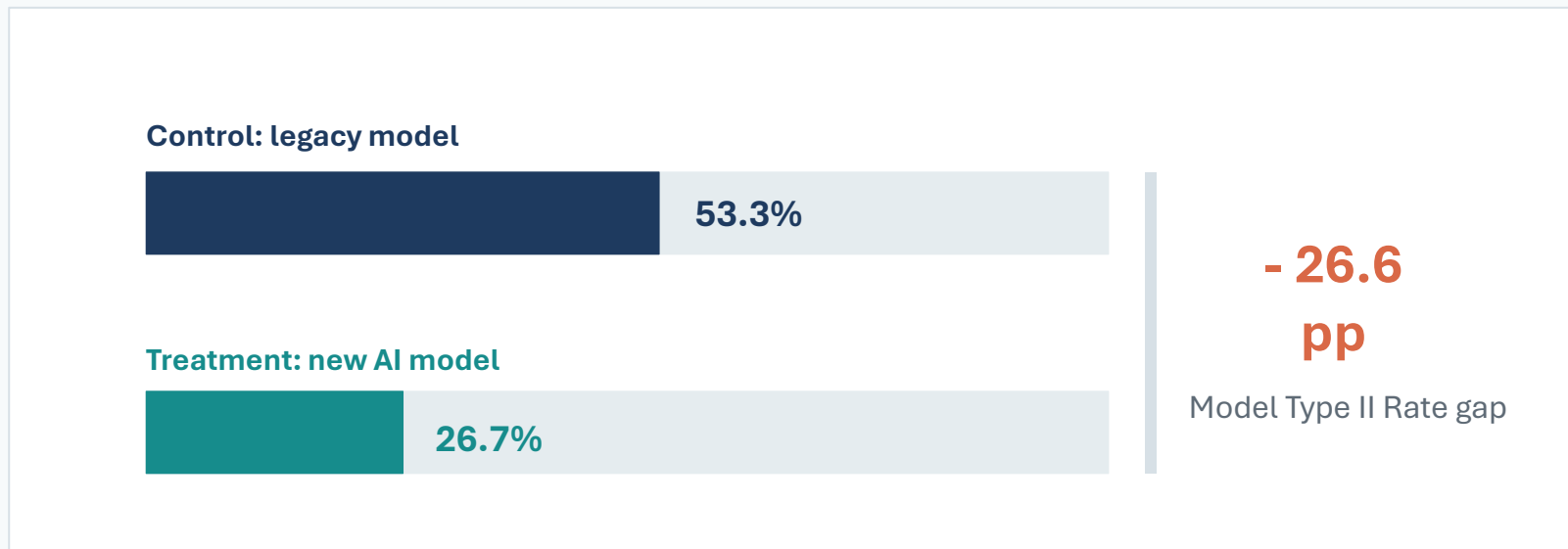
p = 0.0056

Cohen's d = 1.67
large effect size

Better model predictions drove better outcomes.

Improved model accuracy likely contributed to the significant reduction in final Type II errors.

Supporting OEC: Model Type II Rate



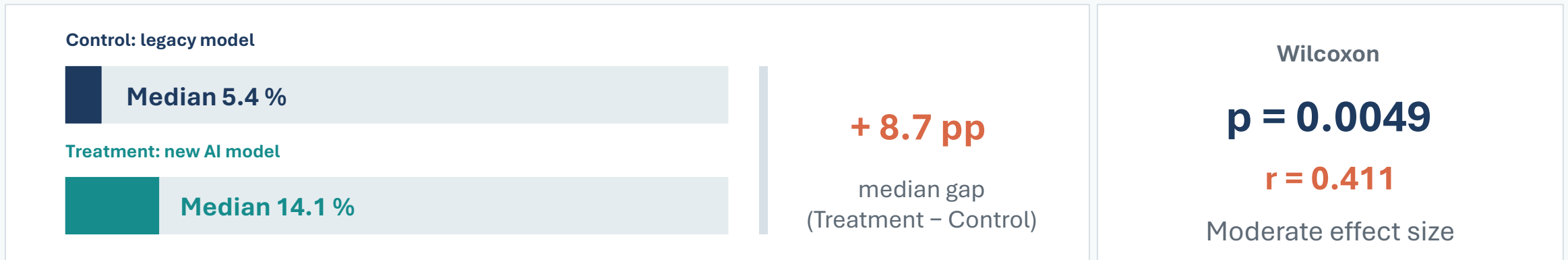
This metric reflects model-level performance, not officer-level variation.

All officers within the same variant share an Model AI Type II Rate — inferential testing is not applicable. Used as a directional signal only.

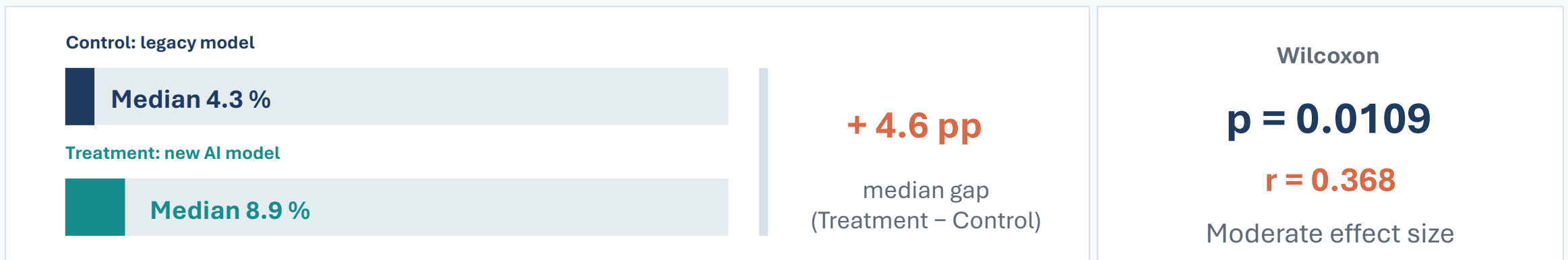
Officers didn't just see better predictions, they trusted and followed them

Both reached statistical significance, reflecting a consistent but moderate change in how officers responded to AI recommendations.

Supporting OEC: Agreement Lift



Supporting OEC: Conflict Reduction



Type II Reduction was not significant—but the result is explainable

The non-significant result is likely driven by sample imbalance and baseline differences, rather than evidence that the new model failed to improve decisions.

Supporting OEC: Type II Reduction

Measures how much each group improved from their own baseline.

Median Type II Reduction



Wilcoxon

p = 0.2559

not significant

r = 0.11

small effect

1 Insufficient statistical power



The small and imbalanced sample reduced the ability to detect modest improvement effects.

2 Baseline differences between groups

Median Initial Type II Rate



Wilcoxon

p = 0.0128

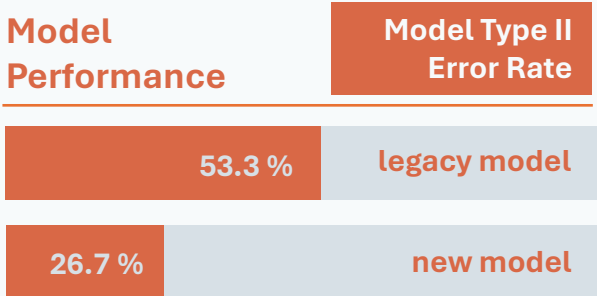
significant difference

The Treatment group started with a lower error rate, leaving less room for improvement.

Comparable reductions therefore reflect a stronger baseline rather than weaker model performance.

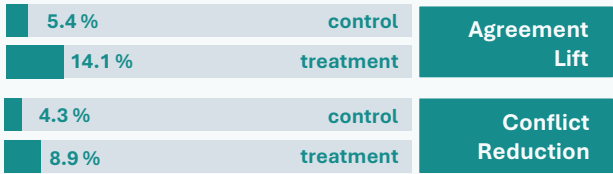
The evidence supports deploying the new AI model.

WHAT THE DATA SHOWS

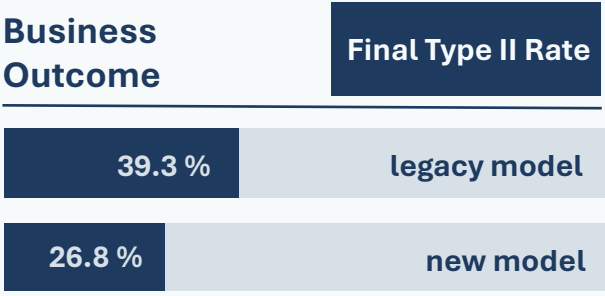


1 New model give better recommendations

Human-Model Collaboration

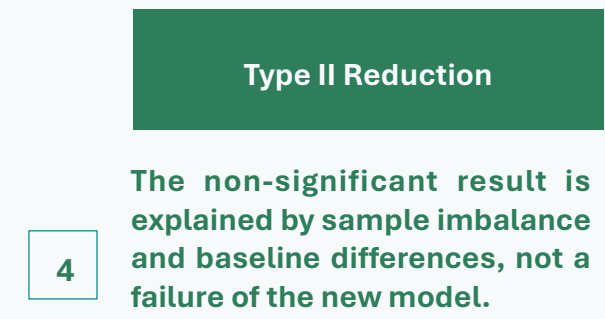


3 Loan officers were significantly more likely to align with the new AI model's recommendations.



2 Deploying the new model led to better final lending decisions

Decision Improvement



4

RECOMMENDED ACTIONS

- 1

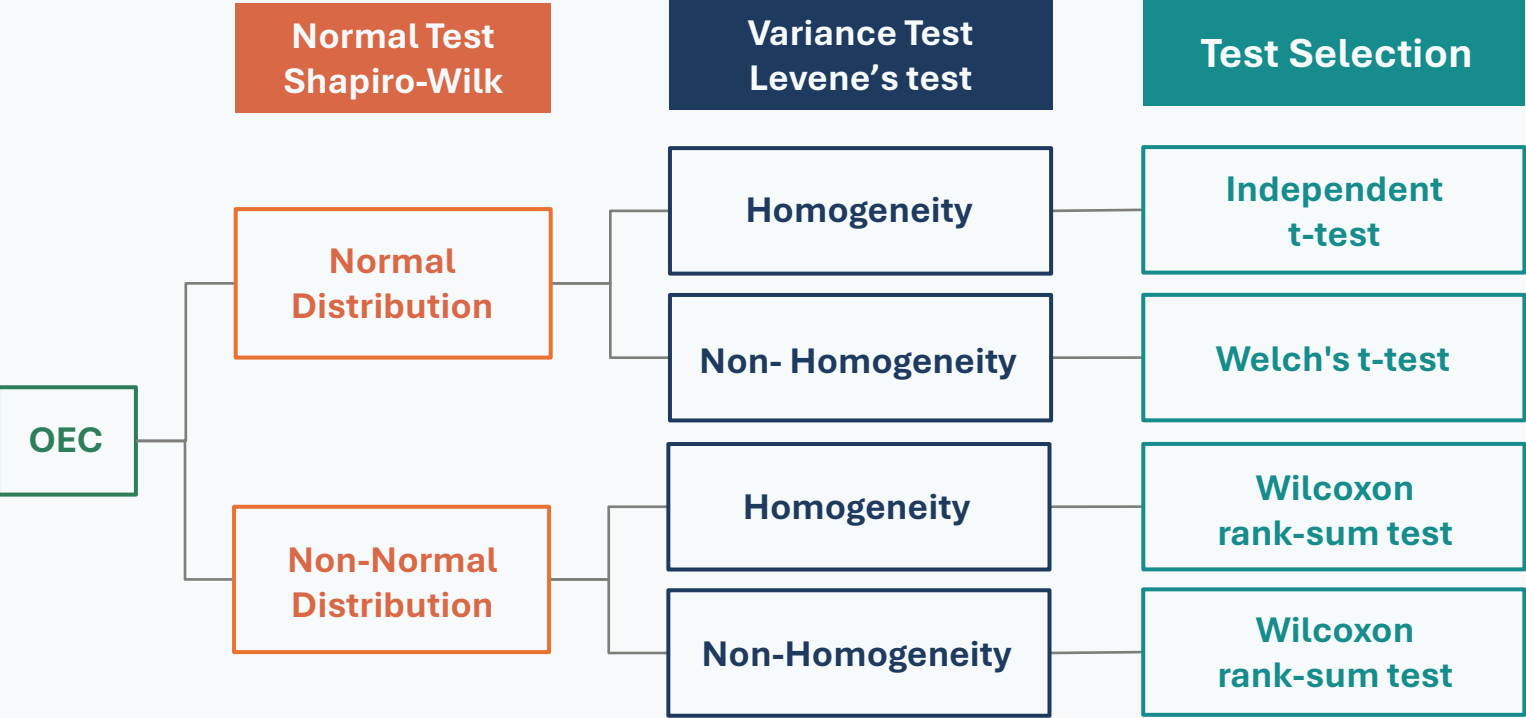
Proceed with controlled rollout
Deploy now to stop legacy model risk exposure and monitor performance continuously.
- 2

Strengthen the experiment
Collect more data during rollout. Target 20–30 per group; consider crossover or matched-pair designs.
- 3

Broaden evaluation
Add Type I Error to capture risk-return trade-off. If case-level data available, run decision-path analysis.

APPENDIX: How were OECs evaluated?

STATISTICAL TEST SELECTION



EFFECT SIZE INTERPRETATION

Measure	Small	Medium	Large
Cohen's d	0.20	0.50	0.80
Wilcoxon r	0.10	0.30	0.50
Cohen's d used with Welch's t-test			
Wilcoxon r used with Wilcoxon rank-sum			

- Welch's T-test: Final Type II Error Rate
- Wilcoxon Rank-sum Test: Type II Reduction, Agreement Lift, Conflict Reduction, Initial Type II Rate